

BRISTOL FILTON, United Kingdom

WMO: 036280

Lat: 51.517N

Lon: 2.583W

Elev: 194

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.4	28.9	21.7	16.4	30.5	23.9	18.2	32.2	51.6	51.3	45.5	48.6	5.4	30	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	79.0	65.5	75.2	63.7	72.1	62.5	67.3	75.3	65.3	72.7	63.8	69.9	9.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.9	70.0	62.7	85.7	67.9	61.2	81.3	66.3	31.8	75.2	30.2	73.0	29.1	70.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.9	25.7	21.9	DB	22.0	84.6	3.1	4.7	19.8	88.0	17.9	90.7	16.2	93.4	13.9	96.8	(4)
(5)			WB	21.3	71.2	3.0	4.4	19.2	74.4	17.4	77.0	15.8	79.4	13.6	82.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	41.8	41.8	44.8	49.6	54.6	60.3	63.4	62.3	58.8	53.4	46.6	42.3	(6)	
	DBStd	9.36	5.94	5.53	5.20	4.79	4.71	4.17	4.43	3.66	4.00	4.91	5.67	6.46	(7)	
	HDD50	1130	258	232	173	61	9	0	0	0	23	127	247		(8)	
	HDD65	4923	720	650	627	461	323	152	86	99	189	360	552	704	(9)	
	CDD50	1752	3	3	11	50	152	309	415	383	265	128	25	8	(10)	
	CDD65	68	0	0	0	0	2	11	36	16	3	0	0	0	(11)	
	CDH74	445	0	0	0	1	18	86	246	75	16	3	0	0	(12)	
	CDH80	74	0	0	0	0	0	15	47	12	0	0	0	0	(13)	
(14) Wind	WSAvg	10.1	11.5	11.4	10.6	10.2	10.7	9.8	9.4	8.8	8.7	9.8	9.6	10.4	(14)	
(15) Precipitation	PrecAvg	36.4	3.9	2.7	2.2	2.4	2.7	2.4	2.5	3.0	2.7	4.0	4.0	3.7	(15)	
	PrecMax	49.5	7.6	6.3	4.2	7.1	5.5	6.6	6.1	5.8	5.6	7.7	8.4	6.9	(16)	
	PrecMin	25.8	0.2	0.1	0.6	0.4	0.4	0.4	0.4	0.2	0.6	1.7	1.5	0.9	(17)	
	PrecStd	5.9	1.8	1.7	1.1	1.5	1.3	1.5	1.3	1.7	1.5	1.7	1.6	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	55.8	63.3	71.4	76.5	81.9	84.7	81.1	76.5	68.0	59.8	57.4	(19)	
		MCWB	52.8	52.7	52.0	57.2	62.9	65.5	66.7	70.7	65.7	61.8	57.0	54.5	(20)	
	2%	DB	53.4	53.1	58.2	65.5	71.3	75.8	80.2	75.3	71.6	63.9	57.7	55.4	(21)	
		MCWB	51.7	50.0	50.4	55.3	60.6	63.3	65.5	65.8	62.3	58.8	55.3	53.5	(22)	
	5%	DB	51.8	51.2	55.0	61.1	67.1	72.5	76.1	71.7	68.6	62.1	56.4	53.5	(23)	
		MCWB	49.9	48.4	49.8	52.5	58.4	62.2	64.1	63.5	60.7	57.9	54.0	51.3	(24)	
	10%	DB	50.2	49.8	52.7	57.8	63.3	68.8	72.2	69.2	66.2	60.5	54.8	51.7	(25)	
		MCWB	48.3	47.6	48.4	50.9	55.9	60.2	62.4	61.6	59.7	56.8	52.5	49.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	53.5	54.0	59.1	65.6	69.3	69.6	74.6	67.1	62.8	58.1	55.2	(27)	
		MCDB	54.5	55.4	58.5	67.6	72.3	78.2	81.0	78.8	73.4	66.4	59.0	56.4	(28)	
	2%	WB	52.1	50.9	52.2	55.9	61.8	65.3	66.6	67.2	64.3	60.4	56.1	53.7	(29)	
		MCDB	53.1	52.3	56.2	63.4	69.5	72.8	77.2	72.6	69.1	62.9	57.2	55.4	(30)	
	5%	WB	50.3	49.4	50.8	53.8	59.3	62.8	65.0	64.8	62.5	59.0	54.4	51.8	(31)	
		MCDB	51.5	50.8	54.3	59.4	65.2	70.0	74.2	69.8	66.8	61.2	55.9	53.3	(32)	
	10%	WB	48.7	47.9	49.2	51.9	57.2	61.1	63.4	63.3	60.9	57.6	52.8	49.9	(33)	
		MCDB	50.1	49.6	52.3	57.0	62.7	67.2	70.3	67.7	64.9	60.0	54.6	51.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.4	12.2	14.1	13.9	13.9	13.8	13.0	13.5	11.0	9.8	9.1	(35)	
		MCDBR	9.3	9.2	15.4	19.5	19.9	20.1	20.6	16.7	16.7	11.7	9.6	9.2	(36)	
	5% WB	MCWBR	9.2	7.5	10.0	11.3	11.1	10.6	9.3	9.4	9.6	7.9	8.3	9.1	(37)	
		MCWBR	9.2	8.3	13.2	16.8	17.6	16.9	17.5	14.9	14.3	10.3	9.4	9.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.327	0.342	0.375	0.396	0.400	0.393	0.390	0.377	0.367	0.353	0.327	0.316	(40)	
		taud	2.355	2.376	2.284	2.242	2.268	2.320	2.333	2.387	2.410	2.422	2.404	2.333	(41)	
	Ebn at Noon		214	246	258	266	271	273	272	270	259	239	213	197	(42)	
		Edh at Noon	20	26	34	39	40	39	37	34	30	24	19	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		248	467	827	1272	1564	1698	1588	1315	990	561	297	193	(44)	
	RadStd		22	57	106	146	79	180	151	116	73	50	29	24	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page